Combustion:
Chemical Kinetics

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Global vs. Elementary Reactions

- Global reactions, e.g.
  - $2H_2 + O_2 = 2H_2O$
  - $CH_4 + 2O_2 = CO_2 + 2H_2O$
- A global reaction provides information on fuel/oxidizer/product ratio, while its reaction rate can hardly be explained by molecular collision theory, e.g. try to explain what happens for:
  - $CH_4 + 2O_2 = CO_2 + 2H_2O$
  - $C_8H_{18} + 12.5O_2 = 8CO_2 + 9H_2O$
- In real life finite-rate chemistry involves elementary steps, e.g.
  - $H_2 + M = H + H + M$
  - $H + O_2 = OH + O$
  - $O + H_2 = OH + H$
  - $OH + H_2 = H_2O + H$
  - …
- Elementary reactions involve intermediate species, which are unstable “molecule segments”, or radicals, generated through bond-breaking by molecular collisions, e.g. H, O, OH, …

Elementary Reaction Rates

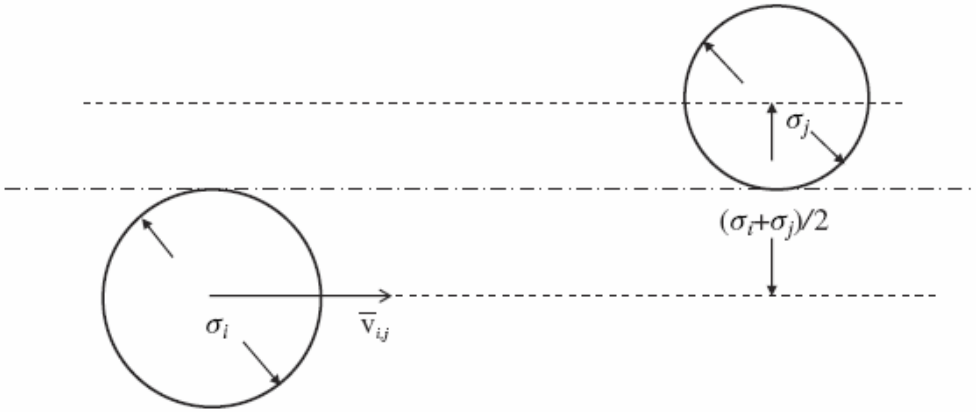
- The reaction rates for elementary reactions are determined by the collision theory

- Consider a bimolecular reaction with two molecules
  - A + B -> products

- Chemical reaction is caused by molecular collisions, therefore the reaction rate depends on
  - The collision frequency per unit volume, Z_{AB}
  - The fraction or probability , P, of collisions that result in chemical bond breaking
  - Empirical factor (steric factor) may apply to match theoretical prediction and experimental results

Collision Frequency of A & B

- Collision frequency is proportional to
  - Molecular size of A and B, measured by the molecular diameter σ_A and σB. Define $2\sigma_{AB} = \sigma_A + \sigma_B$



  - Concentrations of A and B, measured by the number density (#/volume), n_A and n_B
  - Average velocity of A and B, given by the Maxwell distribution,

$$\bar{V}_{AB} = \sqrt{\frac{8kT}{\pi m_{AB}}}, \quad m_{AB} = \frac{m_A m_B}{m_A + m_B}$$

- Putting them together

$$Z_{A,B} = n_A n_B \pi \sigma_{AB}{}^2 \bar{V}_{AB} = n_A n_B \pi \sigma_{AB}{}^2 \sqrt{\frac{8kT}{\pi m_{AB}}}$$

Activation Energy

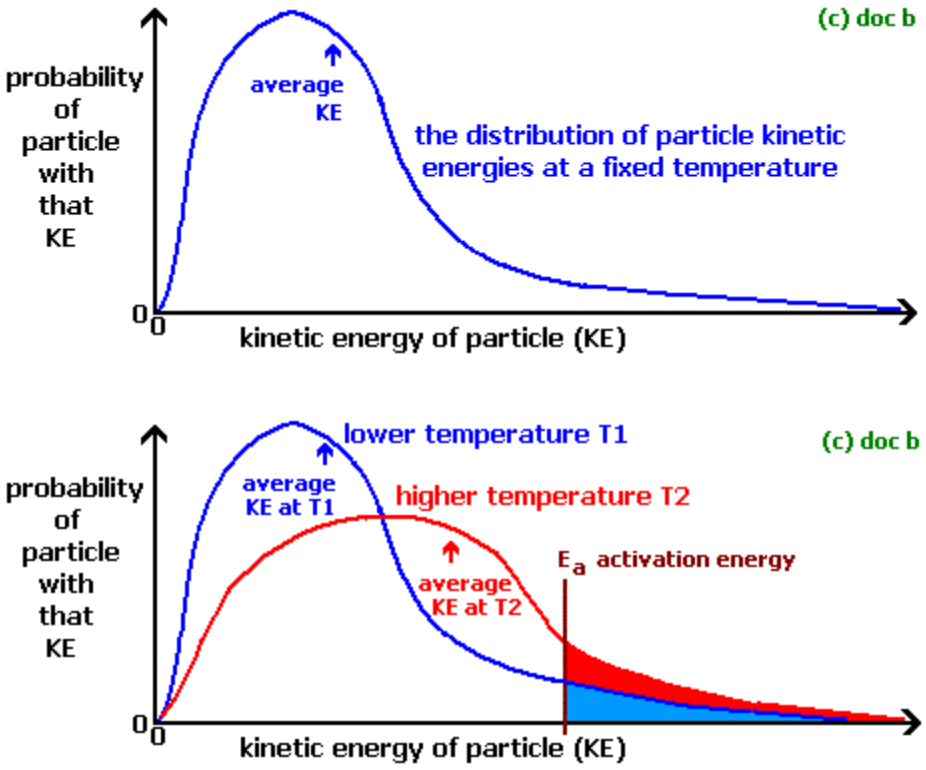
- For a given temperature, the molecules move not in the same speed, but in Maxwell-Boltzmann distribution

- Only those involving a critical kinetic energies (KE) may result in bond-breaking

- This critical KE is called Activation Energy, E_A

- The fraction *P* of collisions with KE>E_A is given by the M-B-distribution

$$P = \exp\left(\frac{-E_A}{R_u T}\right)$$



From: http://www.docbrown.info/page03/ASA2rates.htm

Elementary Reaction Rate

- The reaction rate for A + B -> products:

$$\omega_f = Z_{A,B} P = n_A n_B \pi \sigma_{AB}{}^2 \sqrt{\frac{8kT}{\pi m_{AB}}} \exp\left(\frac{-E_A}{R_u T}\right)$$

$n_A \sim c_A, n_B \sim c_B$ where *c* is the mole concentration

$$\omega_f \sim c_A c_B T^{1/2} \exp\left(\frac{-E_A}{R_u T}\right)$$

- The Arrhenius rate expression, including the steric factor

$$\omega_f = k_f(T) c_A c_B, \quad k_f(T) = A T^n \exp\left(\frac{-E_A}{R_u T}\right)$$

  - *A* is a constant determined by molecular properties of A and B and the steric factor.
  - *A, n, E_A* together determines k_f(T) of the reaction.

The Law of Mass Action

- For a reaction in general form

  $\nu_1'M_1 + \nu_2'M_2 + \ldots + \nu_K'M_K \rightarrow \nu_1''M_1 + \nu_2''M_2 + \ldots + \nu_K''M_K$

- The reaction rate of the reaction is given by

$$\omega_f = k_f(T)\prod_{i=1}^{K} c_i^{\nu_i'} \qquad \omega_r = k_r(T)\prod_{i=1}^{K} c_i^{\nu_i''}$$

- For example:
  - 2A + B -> products $\omega_f = k_f(T)c_A^{\,2}c_B$
  - A + B ⇔ C + D $\omega_f = k_f(T)c_Ac_B \qquad \omega_r = k_r(T)c_Cc_D$

- Calculation of reverse rate $K_C(T) = \dfrac{k_f(T)}{k_r(T)}$

- Recall that the equilibrium constant is

$$K_C(T) \equiv K_p(T)\left(\frac{p^0}{R_u T}\right)^{\sum_{i=1}^{K}\Delta\nu_i} = \exp\left(\frac{-\sum_{i=1}^{K}\Delta\nu_i G_i^0}{R_u T}\right)\left(\frac{p^0}{R_u T}\right)^{\sum_{i=1}^{K}\Delta\nu_i}$$

- The net reaction rate

$$\omega = \omega_f - \omega_r$$

$$= k_f(T)\prod_{i=1}^{K} c_i^{\nu_i'} - k_r(T)\prod_{i=1}^{K} c_i^{\nu_i''} = k_f(T)\prod_{i=1}^{K} c_i^{\nu_i'} - \frac{k_f(T)}{K_C}\prod_{i=1}^{K} c_i^{\nu_i''}$$

- Relation with progress variable: $\frac{d\lambda}{dt} = \omega \cdot V$

- The contribution of the reaction to the rate of change in the concentration of the ith species is: $\frac{dN_i}{dt} = \Delta\nu_i \frac{d\lambda}{dt} = \Delta\nu_i \cdot \omega \cdot V$

Example: One-Step Reaction in Fixed Volume

- A system with fixed volume V, initially filled with N_A, N_B, N_C, N_D moles of A, B, C, and D, respectively, at temperature T
- Consider reaction: A + 2B = C + D

$$\omega = \omega_f - \omega_r \qquad \omega_f = k_f(T) c_A c_B^{\,2} \qquad \omega_r = k_r(T) c_C c_D$$

$$\frac{dN_A}{dt} = \Delta\nu_A \cdot \omega \cdot V \quad \Rightarrow \quad \frac{dc_A}{dt} = \frac{d\left(N_A/V\right)}{dt} = \frac{dN_A}{Vdt} = \Delta\nu_A \cdot \omega = -\omega$$

- Similarly

$$\frac{dc_B}{dt} = \frac{d\left(N_B/V\right)}{dt} = \frac{dN_B}{Vdt} = \Delta\nu_B \cdot \omega = -2\omega$$

$$\frac{dc_C}{dt} = \frac{d(N_C/V)}{dt} = \frac{dN_C}{Vdt} = \Delta\nu_C \cdot \omega = \omega$$

$$\frac{dc_D}{dt} = \frac{d(N_D/V)}{dt} = \frac{dN_D}{Vdt} = \Delta\nu_D \cdot \omega = \omega$$

Rates of Multi-Reaction System

- For a system with the following reactions

$$\nu_{i,1}'M_1 + \nu_{i,2}'M_2 + \ldots + \nu_{i,K}'M_K \Leftrightarrow \nu_{i,1}''M_1 + \nu_{i,2}''M_2 + \ldots + \nu_{i,K}''M_K \qquad i=1,\ldots I$$

- The rate for the ith reaction is:

$$\frac{d\lambda_i}{Vdt} = \omega_i = \omega_{if} - \omega_{ir} = k_{if}(T)\prod_{k=1}^{K} c_i^{\nu_{i,k}'} - k_{ir}(T)\prod_{k=1}^{K} c_i^{\nu_{i,k}''}$$

- The rate of change for the ith species

$$\frac{dN_k}{dt} = \sum_{i=1}^{I}\left(\Delta\nu_{i,k}\frac{d\lambda_i}{dt}\right) = \sum_{i=1}^{I}\left(\Delta\nu_{i,k}\omega_i\right)\cdot V$$

- Again, if V is constant

$$\frac{dc_k}{dt} = \frac{dN_k}{Vdt} = \sum_{i=1}^{I}\left(\Delta\nu_{i,k}\omega_i\right)$$

Species Production Rates In Matrix-Vector Form

- For the kth species $\dfrac{dc_k}{dt} = \sum_{i=1}^{I} \left(\Delta \nu_{i,k} \omega_i \right)$ k=1,2, …, K

- In matrix form: $\dfrac{d\mathbf{c}}{dt} = \mathbf{S} \cdot \boldsymbol{\omega}$

$$\mathbf{c}_{K\times 1} = \begin{pmatrix} c_1 \\ c_2 \\ \cdots \\ c_K \end{pmatrix} \qquad \mathbf{S}_{K\times I} = \begin{pmatrix} \Delta\nu_{1,1} & \Delta\nu_{2,1} & \cdots & \Delta\nu_{I,1} \\ \Delta\nu_{1,2} & \Delta\nu_{2,2} & \cdots & \Delta\nu_{I,2} \\ \cdots & & & \cdots \\ \Delta\nu_{1,K} & \Delta\nu_{2,K} & \cdots & \Delta\nu_{I,K} \end{pmatrix} \qquad \boldsymbol{\omega}_{I\times 1} = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \cdots \\ \omega_I \end{pmatrix}$$

- **S** is the stoichiometric coefficient matrix

Example:

(1) $\mathrm{H_2 + O_2 \rightleftharpoons 2OH}$

(2) $\mathrm{2H_2 + O_2 \rightleftharpoons 2H_2O}$

(3) $\mathrm{H_2 \rightleftharpoons 2H}$

(4) $\mathrm{O_2 \rightleftharpoons 2O}$

(5) $\mathrm{H + OH \rightleftharpoons H_2 + O}$

- Species list:
  - H2, O2, H2O, H, O, OH

$$\mathbf{S}_{6\times5} = \begin{pmatrix} -1 & -2 & -1 & 0 & 1 \\ -1 & -1 & 0 & -1 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & -1 \\ 0 & 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 0 & -1 \end{pmatrix}$$

- The reaction rates

$$\omega_1 = \omega_{1f} - \omega_{1r} = k_{1f} c_{H2} c_{O2} - k_{1r} c_{OH}^{\ 2}$$

$$\omega_2 = \omega_{2f} - \omega_{2r} = k_{2f} c_{H2}^{\ 2} c_{O2} - k_{2r} c_{H2O}^{\ 2}$$

$$\omega_3 = \omega_{3f} - \omega_{3r} = k_{3f} c_{H2} - k_{3r} c_{H}^{\ 2}$$

$$\omega_4 = \omega_{4f} - \omega_{4r} = k_{4f} c_{O2} - k_{4r} c_{O}^{\ 2}$$

$$\omega_5 = \omega_{5f} - \omega_{5r} = k_{5f} c_{H} c_{OH} - k_{5r} c_{H2} c_{O}$$

$$\frac{d\mathbf{c}}{dt} = \mathbf{S} \cdot \boldsymbol{\omega} = \ldots$$

Chain Reactions

- Radicals are important in determining reaction rates
  - e.g. H radical typically controls the oxidation of hydrogen and hydrocarbon
  - Examples of other important radicals OH, O, HCO (for hydrocarbons)

- Based on the creation/consumption of radicals, an elementary reaction can serve for
  - Chain initiation: creates radicals from major species
    e.g. H2 -> H + H
  - Chain propagation: consumes a radical and generates another
    e.g. OH + H2 -> H2O + H
  - Chain branching: generates more radicals than consumed
    e.g. H + O2 -> OH + O, O + H2 -> OH + H
  - Chain termination:
    e.g. H + OH -> H2O

Examples with Chain Reactions

$$M + Br_2 \xrightarrow{k_1} 2Br\bullet + M$$ } Chain initiating step

$$Br\bullet + H_2 \xrightarrow{k_2} HBr + H\bullet$$
$$H\bullet + Br_2 \xrightarrow{k_3} HBr + Br\bullet$$
$$H\bullet + HBr \xrightarrow{k_4} H_2 + Br\bullet$$
} Chain carrying or propagating steps

$$M + 2Br\bullet \xrightarrow{k_5} Br_2 + M$$ } Chain terminating step

$$Cl_2 + M \rightleftarrows Cl + Cl + M$$
$$H_2 + M \rightleftarrows H + H + M$$
$$H + Cl_2 \rightleftarrows HCl + Cl$$
$$Cl + H_2 \rightleftarrows HCl + H$$
$$H + Cl + M \rightleftarrows HCl + M$$

Unimolecular Reactions Involving Third Body: Lindemann mechanism

- For a stable species R with atoms bonded chemically, the chemical species can be activated in a collision with a third body, M, whose kinetic energy passes to R, forming R*

- R* is unstable (vibrating, rotating etc), and the chemical bond is no longer sufficiently strong to keep the atoms in place, R* will then
  - Collide with another third body and pass the kinetic energy back to it, thereby return to the stable state
  - Or, break into other species (products)

$R + M \Leftrightarrow R^* + M$, k_{1f} and k_{1b} are the forward and reverse rate, respectively

$R^* \rightarrow$ products, k_2 is the rate

Quasi Steady State Assumptions

- Example

$$A \xrightarrow{1} B \xrightarrow{1/\varepsilon} C \qquad \tau_{control} \sim O(1)$$

  - Destruction much faster than creation
  - B is a QSS species:

$$\frac{dB}{dt} = A - \frac{B}{\varepsilon} \approx 0 \quad \longrightarrow \quad B \approx A\varepsilon$$

Partial Equilibrium Assumptions

- An example:



$\tau_{control} \sim O(1)$

  - Forward and backward rates are much faster than the net rate
  - Reaction B↔C is in PE:

$$\frac{B}{\varepsilon} - \frac{C}{\varepsilon} \approx 0 \quad \longrightarrow \quad B \approx C$$